

Mathematics A

Unit 2 • Chapter OL-T30 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

7 classified items • 19 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T30 • Expertise Q16+ • 7 items • 19 marks

Chapter	Items	Marks	Starts
01. OL-T30 Functions	7	19	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Functions

Unit 2 • Expertise • 19 marks • 7 item(s)

June 2025 | 4MA1-1H Q16 | 1 marks

Evaluate a function at a numerical input

June 2025 | 4MA1-1H Q16 | 3 marks

Solve a composite function inequality

June 2023 | 4MA1-1H Q17 | 4 marks

Exclude a denominator zero from the domain

January 2022 | 4MA1-2H Q25 | 1 marks

Domain and range swap under inversion

June 2021 | 4MA1-1H Q24 | 2 marks

Evaluate a numerical composite function

June 2021 | 4MA1-1H Q24 | 4 marks

Inverse of a restricted quadratic function

January 2023 | 1H Q18 | 4 marks

Inverse of a composite-defined function

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

$$g(x) = 4x^2$$

(a) Find $f(9)$

(1)

WORKING SPACE

Function substitution

1 marks

Substitute

$$f(9) = 9 - \sqrt{9} = 9 - 3 = 6.$$

Final answer

$$f(9) = 6$$

$$g(x) = 4x^2$$

- (b) Solve $fg(x) < 0$
Show clear algebraic working.

.....
(3)

WORKING SPACE

Composite function inequality**3 marks****Compose the functions rigorously**

$$fg(x) = f(4x^2) = 9 - \sqrt{4x^2} = 9 - 2|x|.$$

Solve the inequality as written

$$9 - 2|x| < 0 \Rightarrow |x| > \frac{9}{2} \Rightarrow x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

Read the domain statement carefully

The printed $x \geq 0$ is the input domain of f . Since $g(x) = 4x^2 \geq 0$ for every real x , it does not remove the negative branch.

If the original variable is additionally restricted

If the original variable is also restricted to $x \geq 0$, then $|x| = x$, so the result reduces to $x > \frac{9}{2}$.

Final answer

$$\text{As written: } x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

$$\text{With an additional } x \geq 0 \text{ restriction: } x > \frac{9}{2}.$$

4MA1-1H Q17

4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of x must be excluded from any domain of g ?

.....
(1)

(b) Solve $gh(x) = 1$

.....
(3)

WORKING SPACE

4MA1-1H Q17

4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

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(a) What value of x must be excluded from any domain of g ?

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(1)

(b) Solve $gh(x) = 1$

.....
(3)

Domain exclusion and composite equation

4 marks

Exclude a zero denominator

$$2x - 5 \neq 0 \Rightarrow x \neq \frac{5}{2}.$$

Form the composite

$$g(h(x)) = \frac{11}{2(x^2 + 4) - 5} = 1.$$

Solve with the stated nonnegative domain

$$2x^2 + 3 = 11 \Rightarrow x^2 = 4, \quad x = 2.$$

Final answer

$$x \neq \frac{5}{2}; \quad x = 2 \text{ for } g(h(x)) = 1$$

4MA1-2H Q25 Q25(b)

1 marks

(b) State the domain of g^{-1}

(1)

WORKING SPACE

4MA1-2H Q25 Q25(b)

1 marks

(b) State the domain of g^{-1}

.....
(1)

Chapter: Functions

1 marks

Q25(b) – Chapter: Functions

Family: Control domain and range • Variant: Domain and range swap under inversion

Method

The square root requires $14-x \geq 0$, so the domain of g^{-1} is $x \leq 14$.

Why: Use the radicand condition.

Final answer

domain: $x \leq 14$

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

.....
(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

Worked solution

June 2021 | Question 24 | 2 marks

Family: Compose functions in the stated order • Variant: Evaluate a numerical composite function

Method / working

1. $g(2) = 7 \times 2 - 6 = 8.$

2. $f(g(2)) = f(8) = 5(8)^2 - 10(8) + 7 = 247.$

Final answer: 247

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

.....
(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

Worked solution

June 2021 | Question 24 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Method / working

1. Write $y = 5x^2 - 10x + 7 = 5(x-1)^2 + 2$ with $x \geq 1$.
 2. Rearrange: $(x-1)^2 = (y-2)/5$.
 3. Use the domain $x \geq 1$, so $x-1$ is the positive square root.
 4. Swap x and y : $f^{-1}(x) = 1 + \sqrt{(x-2)/5}$.
- Final answer: $f^{-1}(x) = 1 + \sqrt{(x-2)/5}$

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2-x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function hg in the form $hg(x) = \dots$
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

(Total for Question 18 is 4 marks)

Answer reference | January 2023 | Question 18

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

18	(a)		$\frac{k}{x}$	1	B1 allow kx^{-1}
	(b)(i)		-46	1	B1 cao
	(ii)	$\frac{3(2-3x^4)}{2-(2-3x^4)}$ or $\frac{6-9x^4}{2-2+3x^4}$ oe or $\frac{6-9x^4}{3x^4}$ oe		2	M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$\frac{2-3x^4}{x^4}$		A1 allow $\frac{2}{x^4} - 3$ oe
					Total 4 marks

Worked solution

January 2023 | Question 18 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

Official final mark-scheme pages are embedded immediately after this question.